

Technical Assessment

DS intern

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Dataset:
147,930

	country	location_name	latitude	longitude	timezone	last_updated_epoch	last_updated	temperature_celsius	temperature_fahr
0	Afghanistan	Kabul	34.52	69.18	Asia/ Kabul	1715849100	2024-05-16 13:15	26.6	
1	Albania	Tirana	41.33	19.82	Europe/ Tirane	1715849100	2024-05-16 10:45	19.0	
2	Algeria	Algiers	36.76	3.05	Africa/ Algiers	1715849100	2024-05-16 09:45	23.0	
3	Andorra	Andorra La Vella	42.50	1.52	Europe/ Andorra	1715849100	2024-05-16 10:45	6.3	
4	Angola	Luanda	-8.84	13.23	Africa/ Luanda	1715849100	2024-05-16 09:45	26.0	

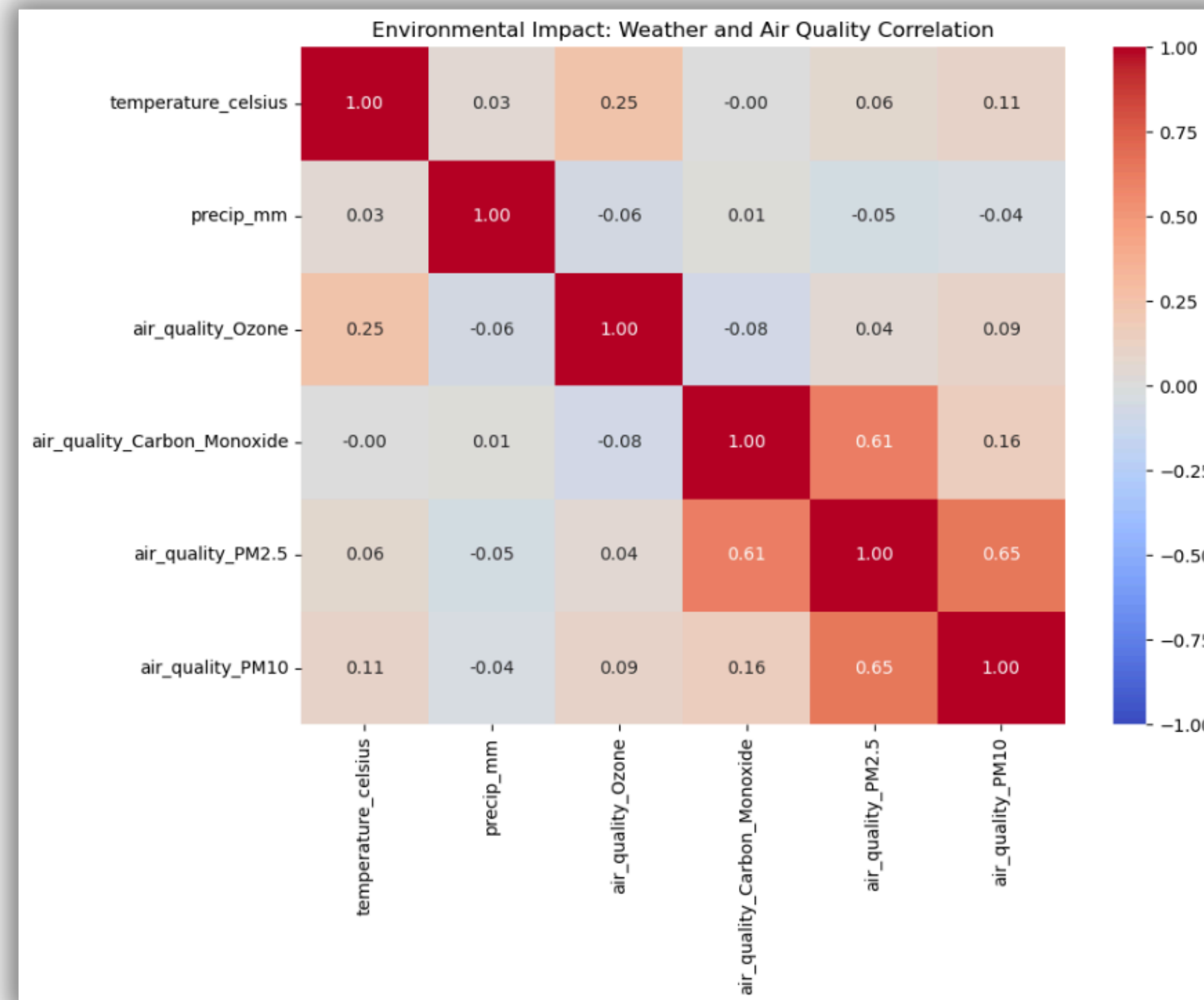
5 rows × 41 columns

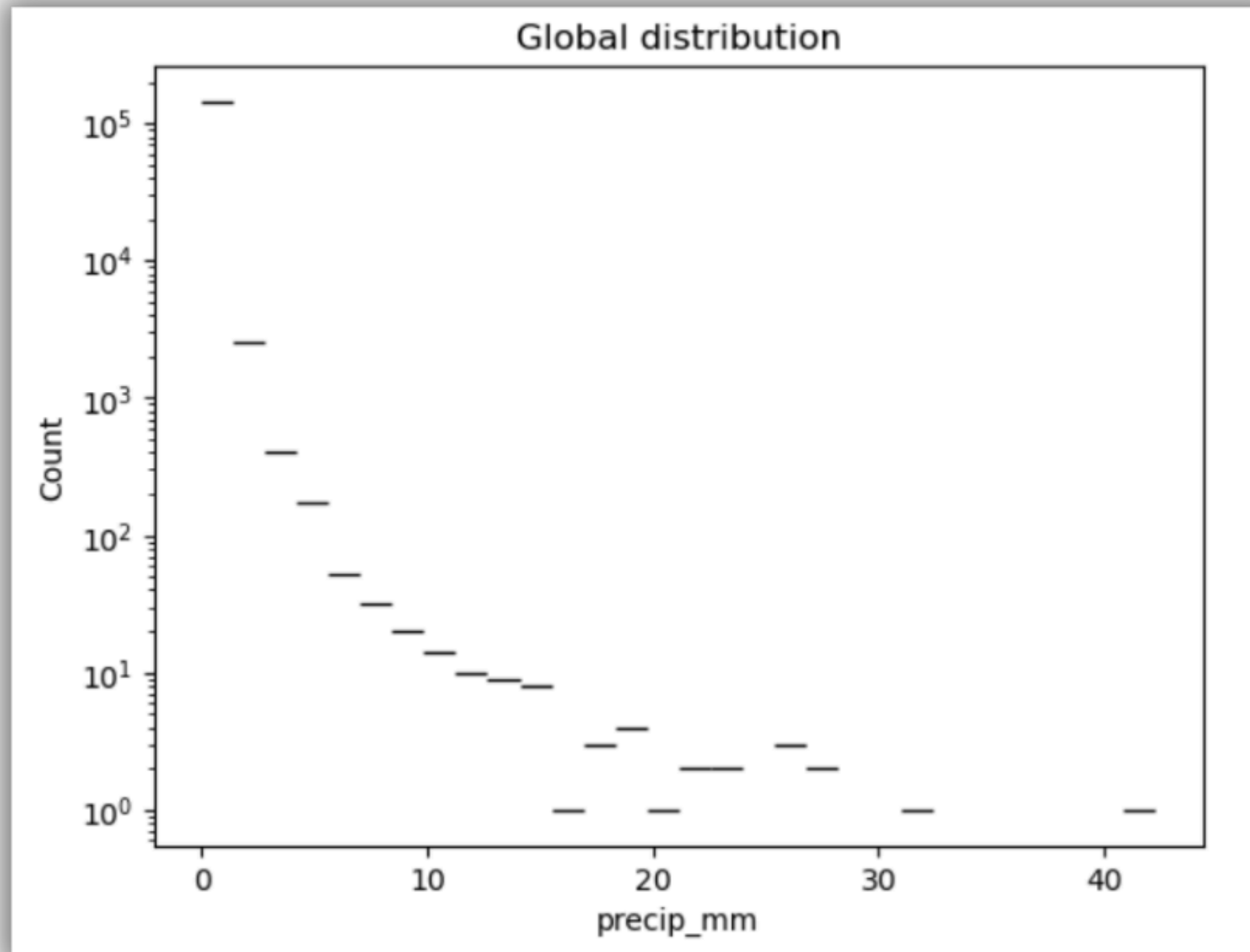
DATA & ANOMALY PROCESSING

- Converted last_updated column to datetime64[ns].
- Cleaned initial temperature extreme outliers (> 50°C).
- Implemented statistical **Interquartile Range (IQR)** method.
- Defined normal global bounds: **-2.10°C to 45.90°C**.
- **Intercepted 2,459 anomalies** (sensor faults & real climate extremes).

EDA & POLLUTION INSIGHTS

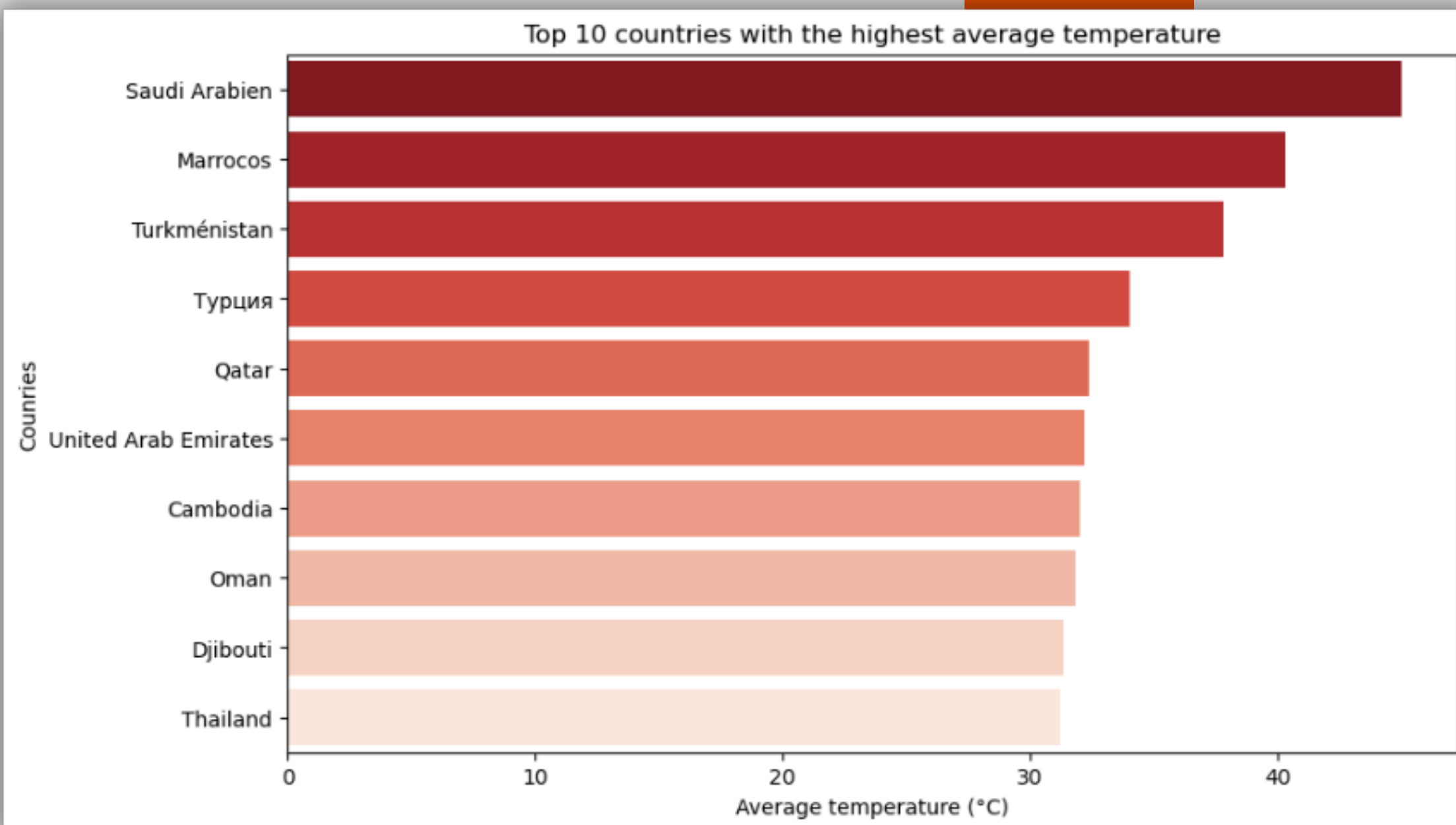
- Tracked global seasonal cycles across the datetime axis.
- Analyzed skewed precipitation metrics using logarithmic scaling.
- Revealed a **0.25** positive correlation between temperature and Ozone.
- Identified strong industrial clustering between **PM2.5** and **PM10 (0.65)**.





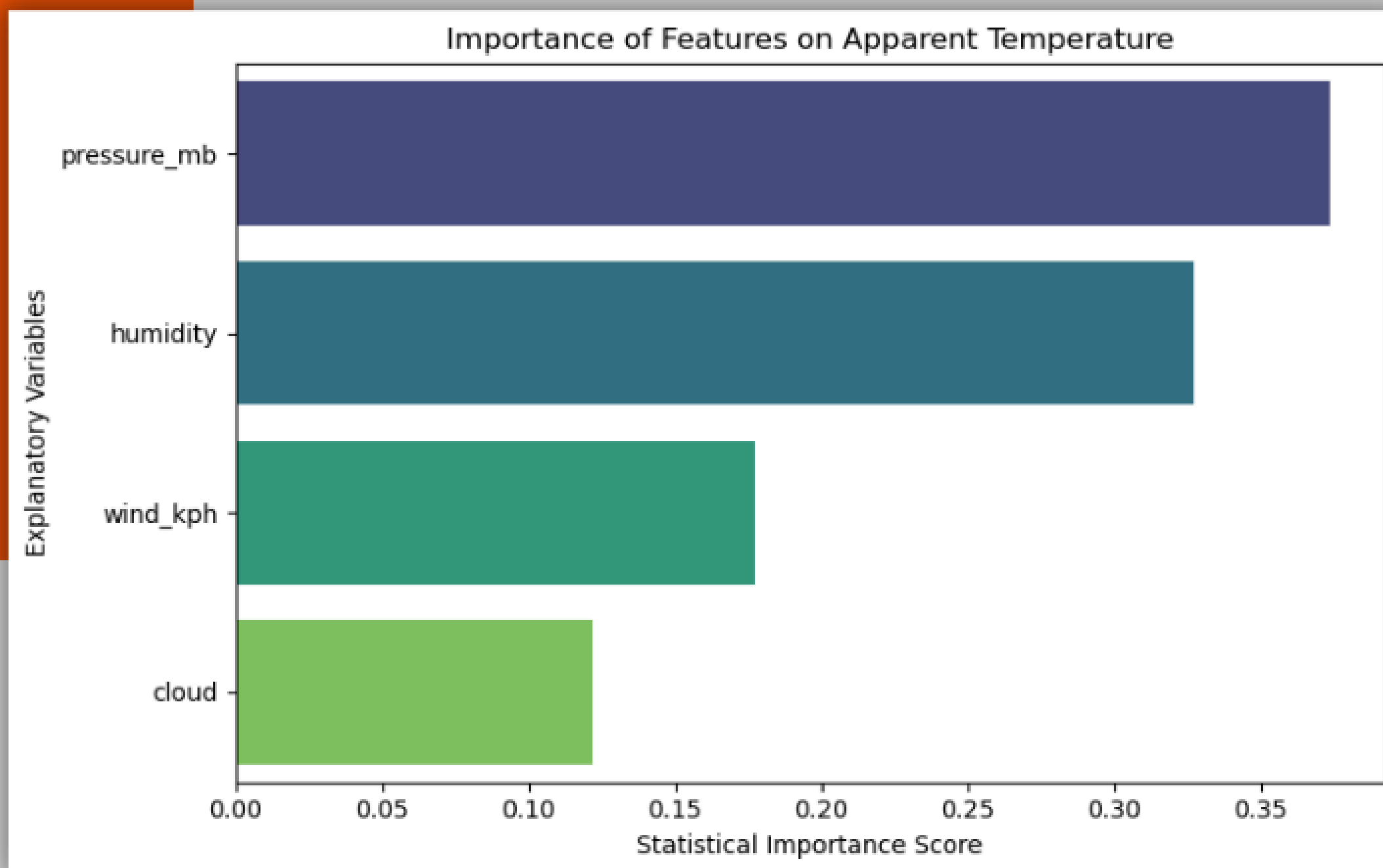
PRECIPITATION DISTRIBUTION

- Analyzed highly skewed climate data using an exponential log-scale.
- Captured high-frequency dry periods concentrated at the **0 mm** baseline.
- Revealed rare but severe precipitation events extending up to **40+ mm**.
- Leveraged robust data transformation to expose true non-linear patterns.



SPATIAL CLIMATE ANALYSIS

- Mapped global climate variations, showing heat dominance in the MENA region.
- Ranked country averages, with Saudi Arabia leading the historical maximums.
- Analyzed geographical pattern variations across diverse longitudinal zones.



FEATURE IMPORTANCE DRIVERS

- Trained a DecisionTreeRegressor to isolate factors driving felt temperature.
- Identified barometric pressure (pressure_mb) and humidity as primary drivers.
- Proved cloud coverage (cloud) holds minimal statistical predictive weight.

FORECASTING & ENSEMBLE

- Enforced chronological data splitting to prevent data leakage.
- Engineered robust lag features using historical time-series metrics.
- Evaluated a linear baseline against a non-linear DecisionTreeRegressor.
- Developed an advanced Arithmetic Ensemble Averaging pipeline.
- Minimized prediction variance to achieve a final 5.08°C MAE.

Forecasting Model	Metric: MAE	Metric: RMSE	Status
Linear Regression (Baseline)	5.17 °C	6.98 °C	Evaluated
Arithmetic Ensemble Model	5.08 °C	Optimized	Best Performance

SUMMARY & DELIVERY

This project delivers a complete, production-ready data pipeline backed by an advanced statistical preprocessing framework. By neutralizing 2,459 climate anomalies via IQR filtering and combining linear baselines with tree regressions, the implemented machine learning ensemble successfully minimized prediction variance to achieve an optimized 5.08°C MAE. The entire architecture is fully documented and open-source for immediate tech team cloning and evaluation.

Video Walkthrough



https://drive.google.com/drive/folders/1O4W89GVl5LTzmV2hu8f9X0AfEanf1ag3?usp=drive_link